

All solutions must clearly show the steps or reasoning you used to arrive at your result. You will lose points for poorly written solutions or incorrect reasoning/explanations; answers given without explanation will not be graded.

Name:
 Section:

- Practice with Laplacians.** (30 points) The (scalar) Laplacian operator ∇^2 is defined independent of coordinate system as the divergence of the gradient: $\nabla^2 f \equiv \nabla \cdot (\nabla f)$. Use the expressions for the gradient and divergence given in cylindrical and spherical coordinates below to compute the Laplacian in these coordinate systems. (The expressions for the Laplacian are readily available in textbooks and on Wikipedia, and you should feel free to use these to check your work, as long as you show the steps in your computation.)

(a) Cylindrical:

$$\nabla f = \frac{\partial f}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial f}{\partial \phi} \hat{\phi} + \frac{\partial f}{\partial z} \hat{z}, \quad \nabla \cdot \mathbf{F} = \frac{1}{\rho} \frac{\partial(\rho F_\rho)}{\partial \rho} + \frac{1}{\rho} \frac{\partial F_\phi}{\partial \phi} + \frac{\partial F_z}{\partial z}$$

We simply take $F_\rho = \frac{\partial f}{\partial \rho}$, $F_\phi = \frac{1}{\rho} \frac{\partial f}{\partial \phi}$, and $F_z = \frac{\partial f}{\partial z}$ from the gradient, and evaluate the divergence:

$$\nabla^2 f = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho} \frac{\partial}{\partial \phi} \left(\frac{1}{\rho} \frac{\partial f}{\partial \phi} \right) + \frac{\partial^2 f}{\partial z^2} = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial f}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 f}{\partial \phi^2} + \frac{\partial^2 f}{\partial z^2}.$$

A good way to check the answer is to count derivatives and use dimensional analysis. The gradient has dimensions of 1/length, so the Laplacian has dimensions of 1/(length)². Since ϕ is a dimensionless angle, its derivatives must come with a factor of 1/ ρ^2 to compensate.

(b) Spherical:

$$\nabla f = \frac{\partial f}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial f}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \hat{\phi}, \quad \nabla \cdot \mathbf{F} = \frac{1}{r^2} \frac{\partial(r^2 F_r)}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial(F_\theta \sin \theta)}{\partial \theta} + \frac{1}{r \sin \theta} \frac{\partial F_\phi}{\partial \phi}$$

Following the same steps as above with $F_r = \frac{\partial f}{\partial r}$, $F_\theta = \frac{1}{r} \frac{\partial f}{\partial \theta}$, and $F_\phi = \frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi}$, we have

$$\begin{aligned} \nabla^2 f &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} \left(\frac{\sin \theta}{r} \frac{\partial f}{\partial \theta} \right) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi} \left(\frac{1}{r \sin \theta} \frac{\partial f}{\partial \phi} \right) \\ &= \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial f}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial f}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 f}{\partial \phi^2}. \end{aligned}$$

Here, we have two dimensionless angular coordinates, which must both come with factors of 1/ r^2 by dimensional analysis.

- Practice with complex numbers.** (30 points)

- (a) Use the polar form to derive formulas for $\sin(3\theta)$ and $\cos(3\theta)$ in terms of $\sin \theta$ and $\cos \theta$.
First, note that $(e^{i\theta})^3 = e^{3i\theta} = \cos(3\theta) + i \sin(3\theta)$. We can also compute the cube as

$$\begin{aligned}(e^{i\theta})^3 &= (\cos \theta + i \sin \theta)^3 = \cos^3 \theta + 3 \cos \theta (-i \sin \theta)^2 + 3 \cos^2 \theta (i \sin \theta) + (i \sin \theta)^3 \\ &= \cos^3 \theta - 3 \cos \theta \sin^2 \theta + 3i \cos^2 \theta \sin \theta - i \sin^3 \theta.\end{aligned}$$

Setting the real part equal to $\cos(3\theta)$ and the imaginary part equal to $\sin(3\theta)$ gives our desired identities:

$$\cos(3\theta) = \cos^3 \theta - 3 \cos \theta \sin^2 \theta; \quad \sin(3\theta) = 3 \cos^2 \theta \sin \theta - \sin^3 \theta.$$

- (b) Compute $\sqrt{i}$ in polar form, and then convert to Cartesian form. (Every complex number has two square roots: find the one which has a phase between 0 and $\pi/2$.)

$$\sqrt{i} = i^{1/2} = (e^{i\pi/2})^{1/2} = e^{i\pi/4} = \frac{\sqrt{2}}{2} + \frac{\sqrt{2}}{2}i$$

- (c) Compute $\frac{3+i}{4+3i}$ in Cartesian form, and then convert to polar form.

First in Cartesian form,

$$\frac{3+i}{4+3i} = \frac{(3+i)(4-3i)}{(4+3i)(4-3i)} = \frac{15-5i}{4^2+3^2} = \frac{3}{5} - \frac{1}{5}i$$

The modulus is $r = \sqrt{(3/5)^2 + (-1/5)^2} = \frac{\sqrt{10}}{5}$, and the phase is $\theta = \tan^{-1}(-1/3)$

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3. **Fourier transforms and derivatives.** (40 points) Consider a vector field $\vec{E}(\vec{x}, t) = \vec{E}_0 e^{i(\vec{k} \cdot \vec{x} - \omega t)}$, where $\vec{E}_0$, $\vec{k}$, and ω are constants. The real part of $\vec{E}$ represents an electromagnetic plane wave (as you've seen in PHYS 212 and Discussion 13).

- (a) By expanding out the dot product in the exponential in components, verify the following three relations:

$$\nabla \cdot \vec{E} = i\vec{k} \cdot \vec{E}, \quad \nabla \times \vec{E} = i\vec{k} \times \vec{E}, \quad \partial_t \vec{E} = -i\omega \vec{E}$$

(Note that we can write the cross product as $\vec{a} \times \vec{b} = \epsilon^{ijk} a_j b_k$; see the index notation review sheet from Lecture 12.) So when we solve an equation with Fourier transformation to get a complex exponential solution, derivatives just turn into multiplication!

$e^{i(\vec{k} \cdot \vec{x} - \omega t)} = e^{ik_x x + ik_y y + ik_z z - i\omega t} = e^{ik_x x} e^{ik_y y} e^{ik_z z} e^{-i\omega t}$. Every spatial derivative just brings down i times the corresponding component of k : for example,

$$\frac{\partial}{\partial x} e^{i(\vec{k} \cdot \vec{x} - \omega t)} = (ik_x) e^{i(\vec{k} \cdot \vec{x} - \omega t)}$$

Therefore, in index notation, $\partial_j \rightarrow ik_j$. The divergence becomes

$$\nabla \cdot \vec{E} = \partial_j E^j = (ik_j) E^j = i\vec{k} \cdot \vec{E},$$

and likewise the curl becomes

$$\nabla \times \vec{E} = \epsilon^{abc} \partial_b E_c = \epsilon^{abc} (ik_b) E_c = i\vec{k} \times \vec{E}.$$

Finally, the time derivative acts on $\partial_t e^{-i\omega t} = (-i\omega) e^{-i\omega t}$, without touching the spatial components, so $\partial_t \vec{E}$ just brings down a factor of $-i\omega$.

- (b) As you saw in Discussion 13, we can combine the wavevector and the frequency into a single 4-vector $k^\mu = (\omega, \vec{k})$, and thus write the exponential as $e^{-ik \cdot x} \equiv e^{-ik_\mu x^\mu}$. Show that

$$\frac{\partial}{\partial x^\nu} (e^{-ik_\mu x^\mu}) = \delta^\mu_\nu (-ik_\mu) e^{-ik \cdot x} = -ik_\nu e^{-ik \cdot x}.$$

So “derivatives turn into multiplication” holds just as well in 4-vector notation, a fact which we will put to use in the final lecture!

This follows exactly the same logic as part (a) as long as we're careful about indices. The partial derivative of x^μ with respect to x^ν is 1 if $\mu = \nu$ and 0 if $\mu \neq \nu$. In other words, $\frac{\partial}{\partial x^\nu} (x^\mu) = \delta^\mu_\nu$. This lets us use the chain rule:

$$\frac{\partial}{\partial x^\nu} (e^{-ik_\mu x^\mu}) = \frac{\partial}{\partial x^\nu} (-ik_\mu x^\mu) e^{-ik \cdot x} = -ik_\mu \delta^\mu_\nu e^{-ik \cdot x}$$

In the second equality we wrote the exponential as $e^{-ik \cdot x}$ without indices to avoid contracting the same dummy index μ twice in the same expression. Contracting the Kronecker delta gives $-ik_\mu \delta^\mu_\nu = -ik_\nu$.